

Aidan Webb

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EDUCATION

University of British Columbia

Bachelor's of Applied Science, Engineering Physics, Dean's List — 85%

Vancouver, BC

2023 — 2027

WORK EXPERIENCE

ScopeSys Microfluidics Instrumentation Engineering (*Co-op*)

May 2026 — Present

- Implemented an open-loop controller in C within a Zephyr firmware stack, with handoff to a tuned PID loop, improving control smoothness ahead of product launch.
- Developed Python-based CLI and GUI tooling for a multi-channel pressure and vacuum control instrument used for nano particle confinement with real time telemetry
- Packaged internal engineering tools into standalone executables using PyInstaller to simplify deployment across laboratory systems.
- Designed precision alignment fixtures and motion-control systems for assembly of multilayer lab-on-chip devices.

PROJECTS

Multi-Agent Simulated Autonomous Drone System

Jan. 2026 - Apr. 2026

Skills Used: ROS1, Gazebo, Python, TensorFlow, OpenCV, PID Control

- Designed a distributed ROS node architecture for multi-agent drone system
- Built multi-drone localization and sensor fusion systems using LiDAR, IMU, and optical flow in ROS/Gazebo.
- Trained and deployed TensorFlow-based image classification and OCR models achieving >98% accuracy.
- Achieved maximum competition score and 2nd place overall out of 15 teams in the ENPH 353 autonomous robotics competition.

EEG Brain-Computer Interface

Jan. 2026 - Ongoing

Skills Used: C/C++, ESP32, Analog Circuit Design, KiCAD, LTspice, UART/I2C

- Designed and validated an EEG acquisition circuit to measure alpha and beta brainwave activity using multi-stage analog amplification and filtering.
- Implemented active filtering and 60 Hz notch suppression circuitry to improve signal integrity for microvolt EEG signals
- Integrated a 16-bit ADC with an ESP32 over I2C for real-time signal acquisition and UART streaming to host software.

Autonomous Collection Robot ([More here!](#))

Jul. 2025 - Aug. 2025

Skills Used: C/C++, ESP32, KiCAD, OnShape/SolidWorks, UART, I2C, ESP-IDF

- Built a fully autonomous robot to sense, locate, and retrieve 7 unique payloads while navigating an obstacle course.
- Designed and fabricated custom ESP32-based PCBs with sensor integration, distributed power regulation, and separated analog/digital signal paths.
- Developed PID motor control and hierarchical state-machine navigation logic, with wireless parameter tuning over WiFi.
- Integrated IMU, Hall-effect, IR, and magnetometer sensors for localization, homing, and autonomous navigation.

16-bit Pipelined RISC CPU ([More here!](#))

Sep. 2024 - Jan. 2025

Skills Used: Verilog, FPGA Development, Quartus, ModelSim, ARM Assembly

- Implemented a custom RISC processor in Verilog with ARM-inspired instruction decoding and memory-mapped I/O.
- Developed the complete processor datapath including ALU, register file, instruction decoder, and pipeline control modules.
- Verified processor correctness using self-checking Verilog testbenches and waveform-based debugging in ModelSim.
- Synthesized and deployed the processor to a DE1-SoC FPGA, executing custom assembly programs and interfacing with onboard hardware switches.

SKILLS

Languages: C/C++, Python, Verilog, ARM Assembly

Embedded/Robotics: ESP32, ROS, RTOS, FPGA Development, PCB Design

Tools: KiCAD, Quartus, ModelSim, Linux, Git, TensorFlow

Fabrication: CAD, Soldering, 3D Printing, Laser Cutting